

Database Systems: Design, Implementation, and Management

Eighth Edition

Relational Algebra

Relational Set Operators

- **Relational algebra**
 - Defines theoretical way of manipulating table contents using relational operators
 - Use of relational algebra operators on existing relations produces new relations
 - UNION INTERSECT
 - DIFFERENCE PRODUCT
 - SELECT PROJECT JOIN DIVIDE

**FIGURE
3.5**

UNION

P_CODE	P_DESCRIPT	PRICE
123456	Flashlight	5.26
123457	Lamp	25.15
123458	Box Fan	10.99
213345	9v battery	1.92
254467	100W bulb	1.47
311452	Powerdrill	34.99

UNION

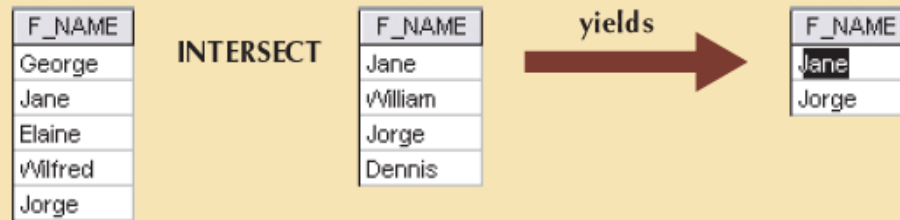
P_CODE	P_DESCRIPT	PRICE
345678	Microwave	160.00
345679	Dishwasher	500.00

yields

P_CODE	P_DESCRIPT	PRICE
123456	Flashlight	5.26
123457	Lamp	25.15
123458	Box Fan	10.99
213345	9v battery	1.92
254467	100W bulb	1.47
311452	Powerdrill	34.99
345678	Microwave	160
345679	Dishwasher	500

**FIGURE
3.6**

INTERSECT



**FIGURE
3.7**

DIFFERENCE



**FIGURE
3.8**

PRODUCT

P_CODE	P_DESCRIPT	PRICE
123456	Flashlight	5.26
123457	Lamp	25.15
123458	Box Fan	10.99
213345	9v battery	1.92
254467	100W bulb	1.47
311452	Powerdrill	34.99

PRODUCT

STORE	aisle	shelf
23	vW	5
24	K	9
25	Z	6

yields

P_CODE	P_DESCRIPT	PRICE	STORE	aisle	shelf
123456	Flashlight	5.26	23	vW	5
123456	Flashlight	5.26	24	K	9
123456	Flashlight	5.26	25	Z	6
123457	Lamp	25.15	23	vW	5
123457	Lamp	25.15	24	K	9
123457	Lamp	25.15	25	Z	6
123458	Box Fan	10.99	23	vW	5
123458	Box Fan	10.99	24	K	9
123458	Box Fan	10.99	25	Z	6
213345	9v battery	1.92	23	vW	5
213345	9v battery	1.92	24	K	9
213345	9v battery	1.92	25	Z	6
311452	Powerdrill	34.99	23	vW	5
311452	Powerdrill	34.99	24	K	9
311452	Powerdrill	34.99	25	Z	6
254467	100W bulb	1.47	23	vW	5
254467	100W bulb	1.47	24	K	9
254467	100W bulb	1.47	25	Z	6

**FIGURE
3.9**

SELECT

Original table

P_CODE	P_DESCRIPTOR	PRICE
123456	Flashlight	5.26
123457	Lamp	25.15
123458	Box Fan	10.99
213345	9v battery	1.92
254467	100W bulb	1.47
311452	Powerdrill	34.99

SELECT ALL yields

New table or list

P_CODE	P_DESCRIPTOR	PRICE
123456	Flashlight	5.26
123457	Lamp	25.15
123458	Box Fan	10.99
213345	9v battery	1.92
254467	100W bulb	1.47
311452	Powerdrill	34.99

SELECT only PRICE less than \$2.00 yields

P_CODE	P_DESCRIPTOR	PRICE
213345	9v battery	1.92
254467	100W bulb	1.47

SELECT only P_CODE = 311452 yields

P_CODE	P_DESCRIPTOR	PRICE
311452	Powerdrill	34.99

**FIGURE
3.10**

PROJECT

Original table

P_CODE	P_DESCRIPT	PRICE
123456	Flashlight	5.26
123457	Lamp	25.15
123458	Box Fan	10.99
213345	9v battery	1.92
254467	100W bulb	1.47
311452	Powerdrill	34.99

PROJECT PRICE yields

New table or list

PRICE
5.26
25.15
10.99
1.92
1.47
34.99

PROJECT P_DESCRIPT and PRICE yields

P_DESCRIPT	PRICE
Flashlight	5.26
Lamp	25.15
Box Fan	10.99
9v battery	1.92
100W bulb	1.47
Powerdrill	34.99

PROJECT P_CODE and PRICE yields

P_CODE	PRICE
123456	5.26
123457	25.15
123458	10.99
213345	1.92
254467	1.47
311452	34.99

**FIGURE
3.11**

Two tables that will be used in join illustrations

Table name: CUSTOMER

CUS_CODE	CUS_LNAME	CUS_ZIP	AGENT_CODE
1132445	Walker	32145	231
1217782	Adares	32145	125
1312243	Rakowski	34129	167
1321242	Rodriguez	37134	125
1542311	Smithson	37134	421
1657399	Vanloo	32145	231

Table name: AGENT

AGENT_CODE	AGENT_PHONE
125	6152439887
167	6153426778
231	6152431124
333	9041234445

Relational Set Operators (continued)

- **Natural Join**
 - Links tables by selecting rows with common values in common attribute(s)
- **Equijoin**
 - Links tables on the basis of an equality condition that compares specified columns
- **Theta join**
 - Any other comparison operator is used
- **Outer join**
 - Matched pairs retained and any unmatched values in other table left null

**FIGURE
3.15**

Left outer join

CUS_CODE	CUS_LNAME	CUS_ZIP	AGENT_CODE	AGENT_PHONE
1217782	Adares	32145	125	6152439887
1321242	Rodriguez	37134	125	6152439887
1312243	Rakowski	34129	167	6153426778
1132445	Walker	32145	231	6152431124
1657399	Vanloo	32145	231	6152431124
1542311	Smithson	37134	421	

**FIGURE
3.16**

Right outer join

CUS_CODE	CUS_LNAME	CUS_ZIP	AGENT_CODE	AGENT_PHONE
1217782	Adares	32145	125	6152439887
1321242	Rodriguez	37134	125	6152439887
1312243	Rakowski	34129	167	6153426778
1132445	Walker	32145	231	6152431124
1657399	Vanloo	32145	231	6152431124
			333	9041234445

**FIGURE
3.17**

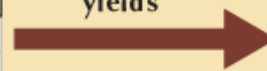
DIVIDE

CODE	LOC
A	5
A	9
A	4
B	5
B	3
C	6
D	7
D	8
E	8

DIVIDE

CODE
A
B

yields



LOC
5

The Data Dictionary and System Catalog

- **Data dictionary**
 - Provides detailed accounting of all tables found within the user/designer-created database
 - Contains (at least) all the attribute names and characteristics for each table in the system
 - Contains metadata: data about data
- **System catalog**
 - Contains metadata
 - Detailed system data dictionary that describes all objects within the database

**TABLE
3.6**

A Sample Data Dictionary

TABLE NAME	ATTRIBUTE NAME	CONTENTS	TYPE	FORMAT	RANGE	REQUIRED	PK OR FK	FK REFERENCED TABLE
CUSTOMER	CUS_CODE	Customer account code	CHAR(5)	99999	10000–99999	Y	PK	AGENT
	CUS_LNAME	Customer last name	VCHAR(20)	Xxxxxxxx	100–999	Y	FK	
	CUS_FNAME	Customer first name	VCHAR(20)	Xxxxxxxx		Y		
	CUS_INITIAL	Customer initial	CHAR(1)	X				
	CUS_RENEW_DATE	Customer insurance renewal date	DATE	dd-mmm-yyyy				
	AGENT_CODE	Agent code	CHAR(3)	999				
AGENT	AGENT_CODE	Agent code	CHAR(3)	999	0.00– 9,999,999.99	Y	PK	
	AGENT_AREACODE	Agent area code	CHAR(3)	999		Y		
	AGENT_PHONE	Agent telephone number	CHAR(8)	999-9999		Y		
	AGENT_LNAME	Agent last name	VCHAR(20)	Xxxxxxxx		Y		
	AGENT_YTD_SLS	Agent year-to-date sales	NUMBER(9,2)	9,999,999.99		Y		

FK = Foreign key
 PK = Primary key
 CHAR = Fixed character length data (1–255 characters)
 VARCHAR = Variable character length data (1–2,000 characters)
 NUMBER = Numeric data (NUMBER(9,2)) is used to specify numbers with two decimal places and up to nine digits, including the decimal places. Some RDBMSs permit the use of a MONEY or CURRENCY data type.

Note: Telephone area codes are always composed of digits 0–9. Because area codes are not used arithmetically, they are most efficiently stored as character data. Also, the area codes are always composed of three digits. Therefore, the area code data type is defined as CHAR(3). On the other hand, names do not conform to some standard length. Therefore, the customer first names are defined as VARCHAR(20), thus indicating that up to 20 characters may be used to store the names. Character data are shown as left-justified.

Relationships within the Relational Database

- 1:M relationship
 - Relational modeling ideal
 - Should be the norm in any relational database design
- 1:1 relationship
 - Should be rare in any relational database design

Relationships within the Relational Database (continued)

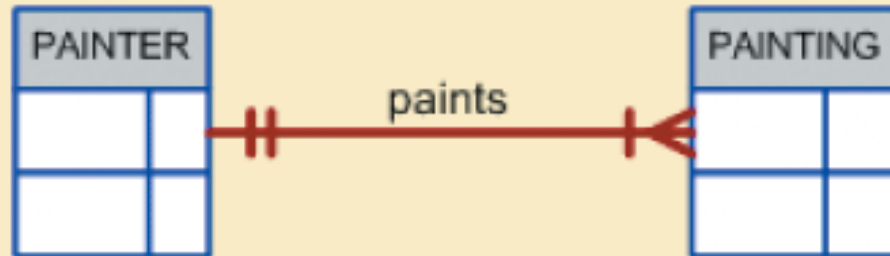
- M:N relationships
 - Cannot be implemented as such in the relational model
 - M:N relationships can be changed into two 1:M relationships

The 1:M Relationship

- Relational database norm
- Found in any database environment

**FIGURE
3.18**

**The 1:M relationship between
PAINTER and PAINTING**



**FIGURE
3.19**

The implemented 1:M relationship between PAINTER and PAINTING

Table name: PAINTER

Primary key: PAINTER_NUM

Foreign key: none

Database name: Ch03_Museum

PAINTER_NUM	PAINTER_LNAME	PAINTER_FNAME	PAINTER_INITIAL
123	Ross	Georgette	P
126	Itero	Julio	G

Table name: PAINTING

Primary key: PAINTING_NUM

Foreign key: PAINTER_NUM

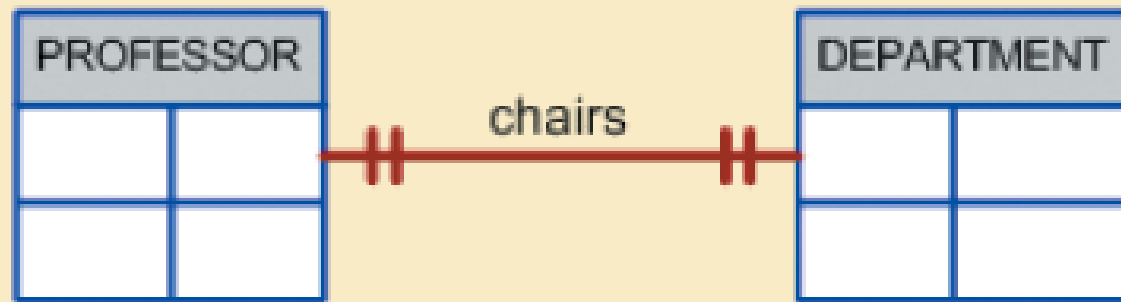
PAINTING_NUM	PAINTING_TITLE	PAINTER_NUM
1338	Dawn Thunder	123
1339	Vanilla Roses To Nowhere	123
1340	Tired Flounders	126
1341	Hasty Exit	123
1342	Plastic Paradise	126

The 1:1 Relationship

- One entity related to only one other entity, and vice versa
- Sometimes means that entity components were not defined properly
- Could indicate that two entities actually belong in the same table
- Certain conditions absolutely require their use

**FIGURE
3.22**

**The 1:1 relationship between
PROFESSOR and DEPARTMENT**

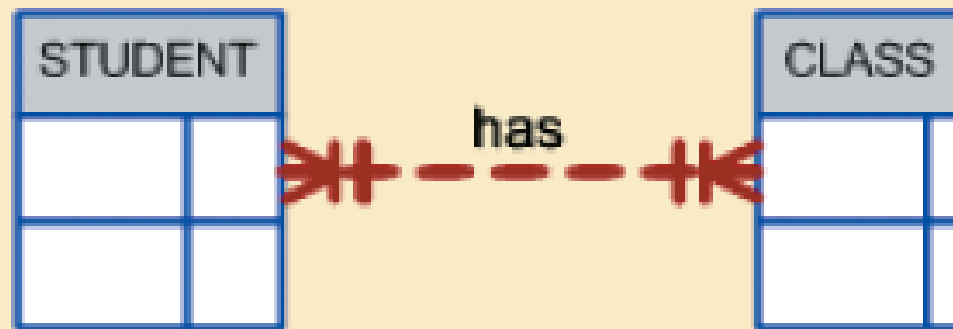


The M:N Relationship

- Implemented by breaking it up to produce a set of 1:M relationships
- Avoid problems inherent to M:N relationship by creating a **composite entity**
 - Includes as foreign keys the primary keys of tables to be linked

**FIGURE
3.24**

**The ERM's M:N relationship
between STUDENT and CLASS**



**FIGURE
3.26**

Converting the M:N relationship into two 1:M relationships

Table name: STUDENT

Primary key: STU_NUM

Foreign key: none

STU_NUM	STU_LNAME
321452	Bowser
324257	Smithson

Database name: Ch03_CollegeTry2

Table name: ENROLL

Primary key: CLASS_CODE + STU_NUM

Foreign key: CLASS_CODE, STU_NUM

CLASS_CODE	STU_NUM	ENROLL_GRADE
10014	321452	C
10014	324257	B
10018	321452	A
10018	324257	B
10021	321452	C
10021	324257	C

Table name: CLASS

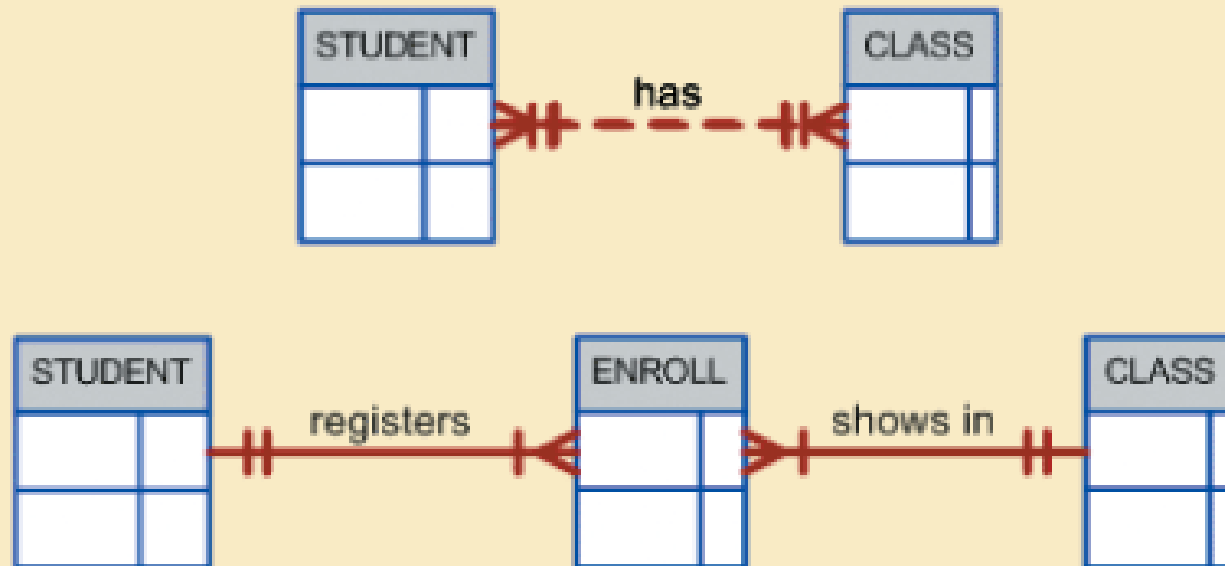
Primary key: CLASS_CODE

Foreign key: CRS_CODE

CLASS_CODE	CRS_CODE	CLASS_SECTION	CLASS_TIME	CLASS_ROOM	PROF_NUM
10014	ACCT-211	3	TTh 2:30-3:45 p.m.	BUS252	342
10018	CIS-220	2	MWF 9:00-9:50 a.m.	KLR211	114
10021	QM-261	1	MWF 8:00-8:50 a.m.	KLR200	114

**FIGURE
3.27**

**Changing the M:N relationship
to two 1:M relationships**



**FIGURE
3.28**

The expanded entity relationship model

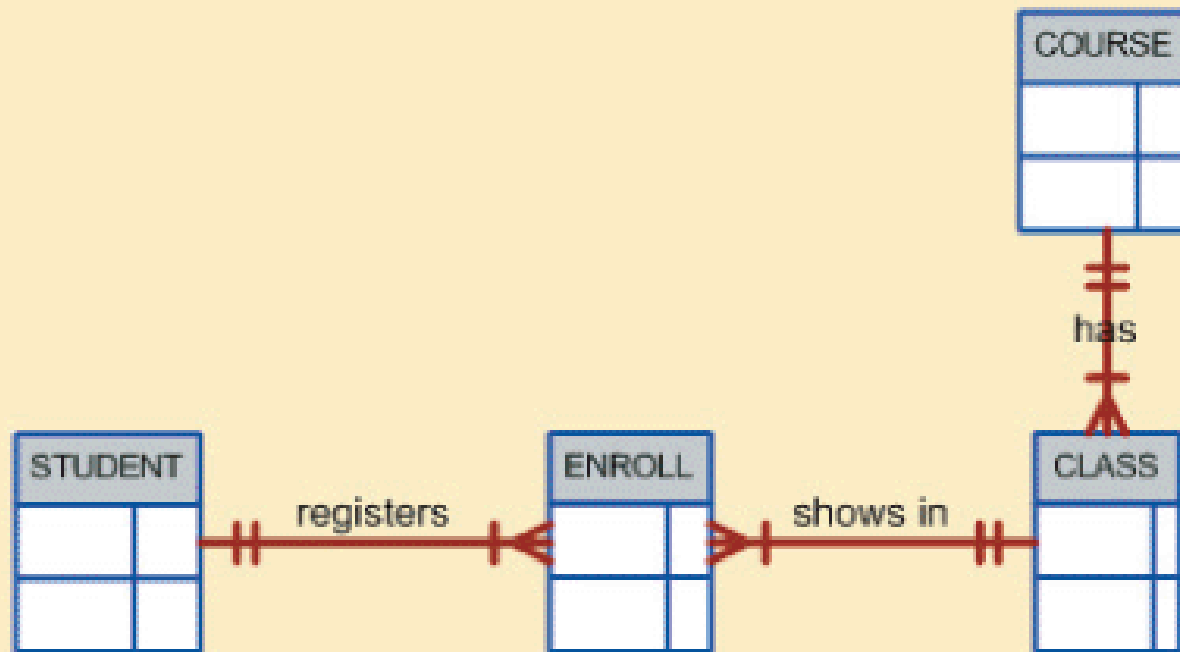
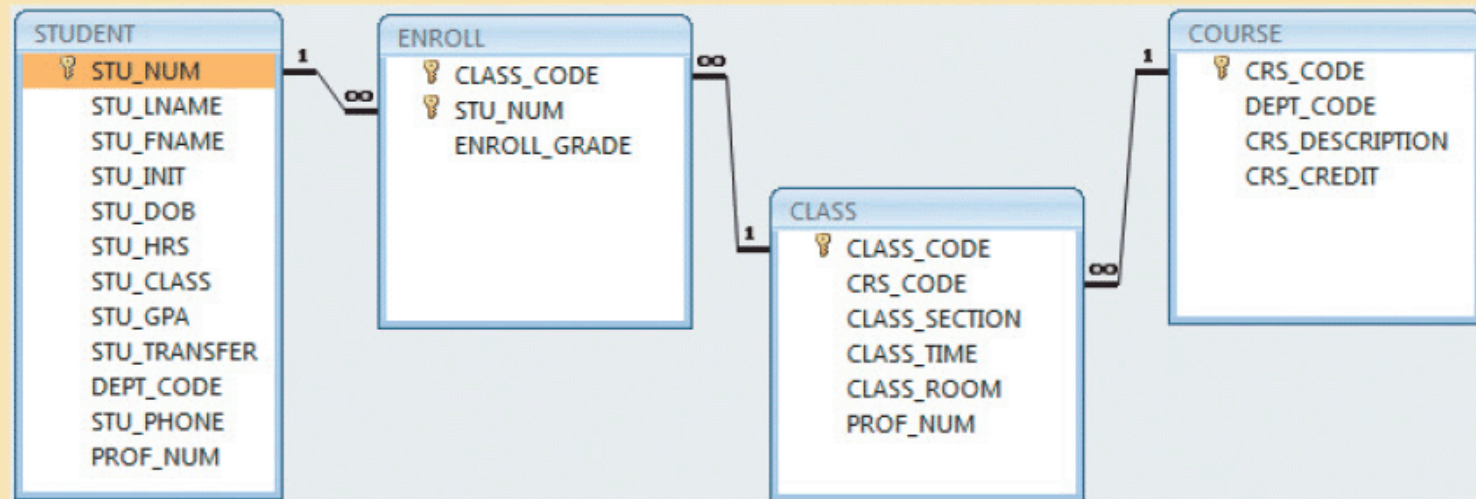


FIGURE 3.29 The relational diagram for the Ch03_TinyCollege database

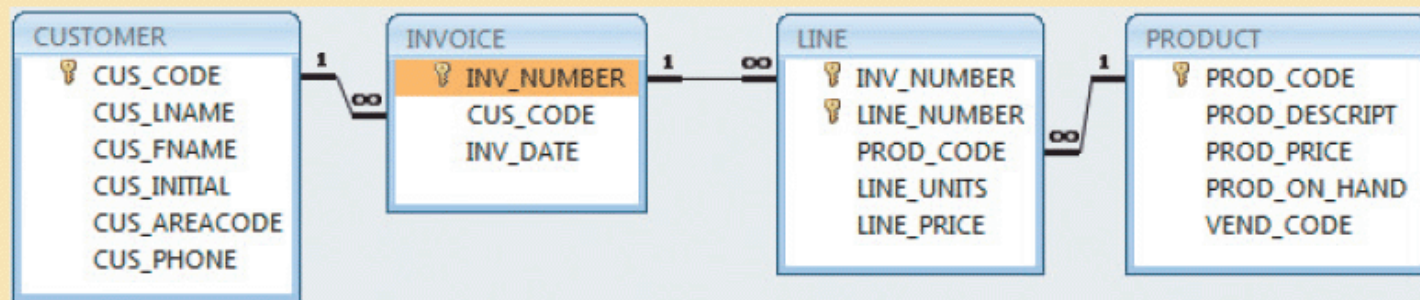


Data Redundancy Revisited

- Data redundancy leads to data anomalies
 - Such anomalies can destroy the effectiveness of the database
- Foreign keys
 - Control data redundancies by using common attributes shared by tables
 - Crucial to exercising data redundancy control
- Sometimes, data redundancy is necessary

**FIGURE
3.31**

The relational diagram for the invoicing system

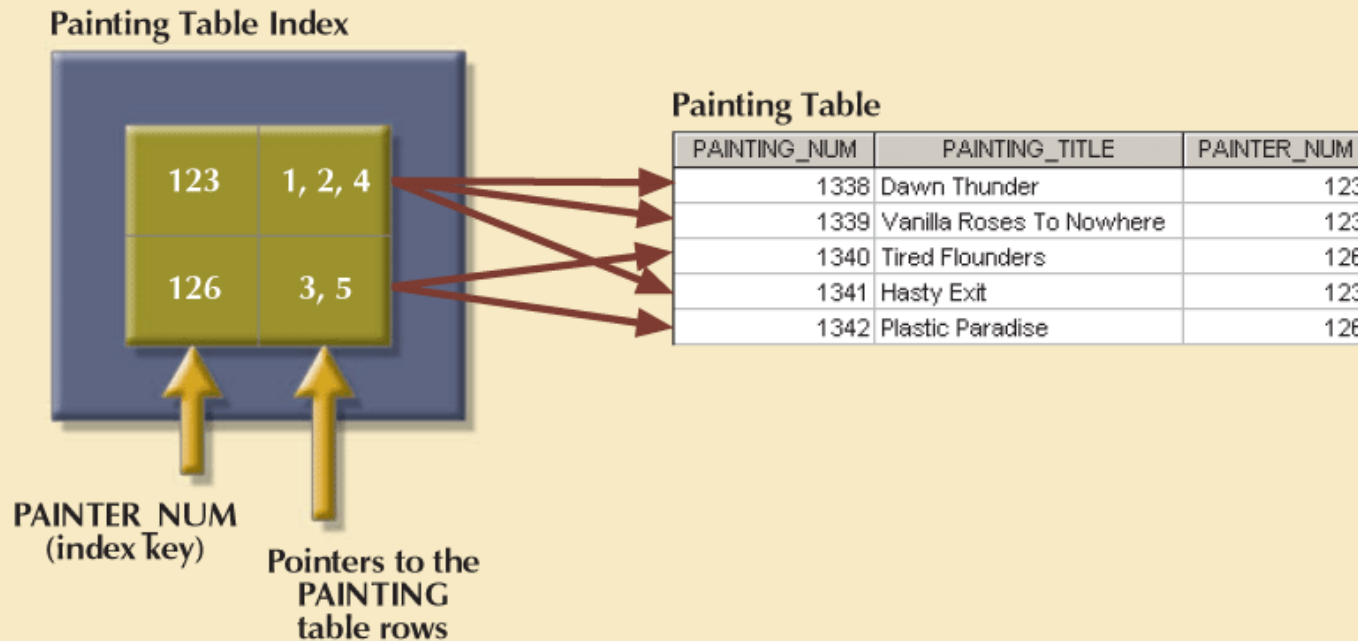


Indexes

- Orderly arrangement to logically access rows in a table
- **Index key**
 - Index's reference point
 - Points to data location identified by the key
- **Unique index**
 - Index in which the index key can have only one pointer value (row) associated with it
- Each index is associated with only one table

**FIGURE
3.32**

Components of an index



Codd's Relational Database Rules

- In 1985, Codd published a list of 12 rules to define a relational database system
 - Products marketed as “relational” that did not meet minimum relational standards
- Even dominant database vendors do not fully support all 12 rules

Summary

- Tables are basic building blocks of a relational database
- Keys are central to the use of relational tables
- Keys define functional dependencies
 - Superkey
 - Candidate key
 - Primary key
 - Secondary key
 - Foreign key

Summary (continued)

- Each table row must have a primary key that uniquely identifies all attributes
- Tables linked by common attributes
- The relational model supports relational algebra functions
 - SELECT, PROJECT, JOIN, INTERSECT UNION, DIFFERENCE, PRODUCT, DIVIDE
- Good design begins by identifying entities, attributes, and relationships
 - 1:1, 1:M, M:N